

Clustering Results

CA Assignment 2 · Clustering Algorithms

Algorithm: Agglomerative Hierarchical Clustering (AHC)

1. Clustering Algorithm: Agglomerative Hierarchical Clustering

Agglomerative Hierarchical Clustering (AHC) is a bottom-up clustering algorithm that builds a hierarchy of nested partitions by iteratively merging the two closest clusters. The process begins with N singleton clusters — one per data point — and terminates when all points belong to a single cluster. A flat partition into K clusters is obtained by cutting the resulting dendrogram at the level that produces K groups. The cut point is selected by maximising the silhouette coefficient over $K = 1, \dots, 10$.

1.1 Algorithm Mechanics

- **Single linkage:** $d(A, B) = \min_{a \in A, b \in B} d(a, b)$. Prone to chaining; produces elongated clusters.
- **Average linkage:** $d(A, B) = \frac{1}{|A||B|} \sum_{a \in A} \sum_{b \in B} d(a, b)$. Less prone to chaining but susceptible on imbalanced corpora.
- **Ward linkage (used in this work):** merges the pair (A, B) minimising the increase in total within-cluster sum of squares. Produces compact, balanced, well-separated clusters.

1.2 Why AHC?

- **No assumption about cluster shape.** Unlike K -means, AHC recovers clusters of arbitrary shape and size, accommodating the strong size imbalance in this dataset (Shakespeare $\approx 46\%$, modern English $\approx 49\%$, Notre Dame sub-corpora $< 5\%$).
- **Deterministic output.** AHC produces the same result on every run, unlike K -means, which depends on random centroid initialisation.
- **No need to pre-specify K .** AHC builds the complete dendrogram in a single pass; the optimal K is selected post-hoc by silhouette analysis.
- **Literature support.** AHC is a well-validated algorithm for text clustering with dense embeddings (Petukhova et al., 2024).

2. Configuration: Ward Linkage and Euclidean Distance

2.1 Ward Linkage

Ward linkage (Ward, 1963) selects the merge that minimises the increase in total within-cluster sum of squares (WCSS):

$$\Delta(A, B) = \frac{|A||B|}{|A| + |B|} \|\mu_A - \mu_B\|^2$$

Ward is the only linkage criterion that directly optimises cluster compactness at every merge step, and is robust against the chaining problem.

2.2 Euclidean Distance

Ward linkage is mathematically defined *only* for Euclidean distance. The feature matrix passed to AHC is the 5-dimensional UMAP output, which lies in an unconstrained Euclidean space. Euclidean distance in this space is therefore both the correct and the only valid metric for Ward-linkage AHC.

Furthermore, L2 normalisation was applied to the raw embeddings before PCA, and UMAP’s neighbourhood graph was built using cosine distance. For unit vectors, Euclidean and cosine distances are monotonically equivalent:

$$d_{\text{Euclidean}}(\hat{\mathbf{a}}, \hat{\mathbf{b}})^2 = 2(1 - \cos(\hat{\mathbf{a}}, \hat{\mathbf{b}}))$$

This ensures that Euclidean distances in the UMAP output space remain consistent with the cosine-based semantic geometry of the embedding model.

3. Silhouette Analysis

The silhouette coefficient is an internal cluster validity index (Rousseeuw, 1987):

$$\begin{aligned} a(i) &= \text{mean intra-cluster Euclidean distance from } i \\ b(i) &= \text{mean Euclidean distance from } i \text{ to the nearest other cluster} \\ s(i) &= \frac{b(i) - a(i)}{\max(a(i), b(i))} \in [-1, +1] \end{aligned}$$

$$S = \frac{1}{N} \sum_{i=1}^N s(i)$$

Silhouette is computed using Euclidean distance in the same 5-dimensional UMAP space as clustering — the only self-consistent choice. Per recent literature (PMC, 2025): $S > 0.70$ = strong, 0.50 – 0.70 = acceptable, 0.25 – 0.50 = weak.

3.1 Results for $K = 1$ to 10

4. Optimal Value of K and Justification

4.1 Quantitative Justification: Silhouette Analysis

The silhouette analysis identifies $\mathbf{K} = 4$ as the optimal number of clusters, with a peak coefficient of **0.758182**. This score lies firmly in the “strong” range ($S > 0.70$; PMC, 2025).

The curve has a clear structural signature. From $K = 2$ to $K = 6$, scores remain close together in the range $[0.742, 0.758]$, indicating that several valid partitions exist in this region. $K = 4$ achieves the peak. At $K = 7$ there is a sharp drop to 0.571 , followed by a further collapse to 0.352 at $K = 8$. This cliff-edge pattern is a strong quantitative signal: $K \leq 6$ corresponds to genuine

K	Silhouette Coefficient	Notes
1	N/A	Undefined — requires at least 2 clusters
2	0.754644	
3	0.742843	
4	0.758182	Global optimum (peak)
5	0.749800	
6	0.750828	
7	0.571270	Sharp drop
8	0.351745	
9	0.356212	
10	0.390376	

Table 1: Silhouette coefficients for AHC (Euclidean, Ward) over $K = 1$ to 10. $K = 4$ is highlighted as the global optimum.

thematic structure in the data, while $K \geq 7$ begins fracturing coherent clusters arbitrarily. The peak at $K = 4$ identifies the most compact and well-separated partition within this valid range.

4.2 Qualitative Justification: Semantic Cluster Interpretation

The four clusters produced at $K = 4$ correspond to four distinct thematic groups in the dataset:

4.3 Cluster-by-Cluster Analysis

Cluster 3 — Shakespeare (811 documents). The largest cluster captures all Shakespearean dialogue from *Coriolanus*. Its separation from every other cluster is total: Early Modern English vocabulary, syntactic structures (e.g., inverted word order, subjunctive mood), and the character prefix format are unique in the dataset. `all-mpnet-base-v2` embeddings place Shakespearean text in a region of $\mathbb{R}^{768}$ that is maximally distant from any contemporary prose.

Cluster 2 — Contemporary English Prose (861 documents). The second-largest cluster contains informal restaurant and service reviews (Yelp-style) alongside formal news reports (CNN-style). Despite differences in formality, both sub-corpora share modern vocabulary, contemporary syntax, and prose sentence structure — the features that the embedding model weighs most heavily. The cluster boundary between contemporary reviews and news is insufficient to form a separable geometric boundary in the UMAP space, consistent with the observation that only 2 CNN sentences exist in the dataset.

Clusters 1 and 4 — Notre Dame (73 + 15 documents). The Wikipedia Notre Dame sub-corpus (80 rows) is detected as two distinct clusters. Cluster 1 contains the general institutional content (student organisations, media, campus history), while Cluster 4 contains exclusively the sentences describing religious and iconic architectural landmarks (Gold Dome, Grotto, Basilica, Marian statues). This split reflects a genuine semantic boundary within the Notre Dame corpus: the landmark sentences form a highly cohesive, repetitive group (each of the 3 unique sentences in Cluster 4 appears 5 times) whose local UMAP neighbourhood is almost entirely self-referential, producing a tighter, more isolated manifold region than the broader general institutional content of Cluster 1.

Cluster	Size	Domain	Linguistic Characteristics
1	73 (4.1%)	Notre Dame — General	Encyclopaedic prose about Notre Dame University: student media outlets, campus organisations, and institutional history. Formal, third-person, contemporary academic vocabulary.
2	861 (48.9%)	Contemporary English Prose	Predominantly casual restaurant and service reviews (Yelp-style), together with a small number of formal news reports (CNN-style). All share modern vocabulary, contemporary syntax, and first- or third-person prose structure.
3	811 (46.1%)	Shakespeare (Coriolanus)	Early Modern English: play dialogue from Coriolanus. Archaic vocabulary, poetic metre, character prefixes (e.g., CORIOLANUS:, AUFIDIUS:), iambic pentameter constructions.
4	15 (0.9%)	Notre Dame — Religious Landmarks	A highly cohesive sub-cluster of sentences describing iconic religious and architectural features: the Gold Dome, the Grotto, the Basilica, and statues of the Virgin Mary and Christ. Characterised by repeated religious proper nouns and formulaic descriptive prose.

Table 2: Semantic interpretation of the four clusters produced by AHC at $K = 4$.

4.4 Role of Duplicate Retention

The Notre Dame sub-corpus consists of 18 unique sentences, each repeated 4–5 times (80 rows total). Without these repetitions, the 18 unique sentences would represent only 1.1% of the data. At $n_neighbors=10$, each Notre Dame sentence would not find enough Notre Dame neighbours to form a cohesive sub-graph, and the sub-corpus would dissolve into the broader modern English cluster.

By retaining duplicates, the neighbourhood graph of each Notre Dame sentence is dominated by other Notre Dame sentences. UMAP projects this dense, self-referential sub-graph into a compact, isolated region of the 5-dimensional output space. Ward-linkage AHC then cleanly separates this region from the surrounding modern English manifold. The cluster boundary is robust: the silhouette score at $K = 4$ (0.758) exceeds that at $K = 3$ (0.743), confirming that the Notre Dame cluster is genuinely well-separated and not an artefact.

4.5 Contextualisation Against the Literature

Silhouette scores computed in a well-formed 5-dimensional UMAP space (BERTopic standard: `metric='cosine'`, `min_dist=0.0`, `n_components=5`; Grootendorst, 2022) are structurally higher than scores computed in raw embedding or PCA space, because UMAP sharpens cluster boundaries in the projected space. The score of 0.758182 at $K = 4$ is self-consistent and directly meaningful as a measure of cluster separation in the representation the algorithm operated on, and falls in the “strong” range of the current literature (PMC, 2025).

5. Additional Considerations

5.1 Duplicate Handling

All 1,760 rows — including the 64 duplicate sentences — are retained throughout preprocessing, embedding, and clustering. `label.txt` therefore contains exactly 1,760 entries, one per original dataset row.

5.2 Embedding Model

`all-mpnet-base-v2` (768-dim, 110 M parameters, 1 B training pairs) is the highest-quality standard sentence transformer on the Sentence Embeddings Benchmark. It runs fully locally with no cloud API dependency. Petukhova et al. (2024) confirm that BERT-family models represent the best open-source option for text clustering.

References:

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